

DESIGN AND DEVELOPMENT OF A CONTEXT-AWARE SENTIMENT ANALYSIS SYSTEM USING MULTI-LEVEL NLP

GROUP 4 – ASPECT-BASED SENTIMENT ANALYSIS (ABSA) MULTI-LEVEL NLP APPROACH

Academic Project Report

Same Problem Statement as Group 1; distinct core approach: Aspect-Based Sentiment Analysis

Group 4 keeps the same customer-review problem and five-unit NLP coverage while using aspect-based modelling as the sentiment core.

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1. Problem Formulation

Online customer reviews frequently contain multiple opinions within the same review. A review may contain positive opinions about one product aspect and negative opinions about another aspect. Simple sentiment analysis systems that classify text by counting positive and negative words may therefore produce incorrect results.

The problem considered in this assignment is the development of a Python-based context-aware sentiment analysis system that classifies a customer review as Positive, Negative, or Neutral by considering linguistic information at multiple levels.

The system must analyze morphology, N-grams, POS information, syntactic structure, negation, semantic meaning, word sense, reference expressions, discourse relations, product aspects, and overall sentiment.

Customer review used for analysis:

“I was expecting the phone to be amazing. The display is bright and the camera takes excellent pictures, but the battery is disappointing. It does not last long, and I am unhappy with it. However, the service team replaced it quickly, which was impressive. Overall, I am satisfied with the phone.”

The proposed Group 4 solution uses Aspect-Based Sentiment Analysis (ABSA) as the core sentiment method. Instead of assigning one undifferentiated sentiment to the entire review, the system extracts product aspects, links opinions to each aspect, adjusts those opinions using negation and discourse context, and then aggregates aspect-level results to determine the overall sentiment.

Input

A natural-language customer review containing opinions about one or more product aspects.

Output

- Morphological analysis
- N-gram representations
- POS information
- Syntactic/CFG analysis
- Negation detection
- Semantic representation
- Word Sense Disambiguation
- Reference resolution
- Discourse relations
- Aspect identification
- Aspect-level sentiment
- Aspect-weighted overall sentiment score
- Final sentiment classification
- Natural-language explanation

Problem

How can the sentiment of a customer review be correctly determined when positive and negative opinions, negation, contrast, references, and multiple product aspects occur in the same text, while explicitly preserving which opinion belongs to which product aspect?

Why Keyword-Based Analysis Is Insufficient

- “does not last long” expresses a negative opinion through negation.
- “but” changes the discourse relationship between camera/display and battery.
- “However” introduces a contrasting positive opinion about service.
- “it” refers to different entities depending on context.
- “Overall” indicates a global/final opinion.
- Different aspects of the same product have different sentiment polarities.
- A sentiment word should not be separated from the aspect or event it describes.

Therefore, Group 4 uses ABSA to preserve aspect-opinion relationships and uses multi-level linguistic evidence to refine each aspect-level decision.

2. Objectives

1. To develop an aspect-based multi-level NLP system for context-aware sentiment analysis.
2. To preprocess and tokenize customer reviews.
3. To perform morphological analysis and stemming.
4. To generate unigram, bigram, and trigram representations as supporting features.
5. To perform POS-based linguistic analysis for aspect and opinion extraction.
6. To construct CFG rules for syntactic analysis.
7. To explicitly detect and handle negation.
8. To represent selected statements using First-Order Predicate Calculus.
9. To perform contextual Word Sense Disambiguation.
10. To resolve references such as “it” and “which”.
11. To identify discourse relations such as contrast and conclusion.
12. To identify product aspects such as display, camera, battery, service, and phone.
13. To identify aspect-opinion pairs and assign sentiment to each aspect.
14. To use negation as an aspect-level adjustment rather than a global keyword flip.
15. To apply discourse weighting to aspect-level sentiment.
16. To combine local aspect evidence into an overall sentiment decision.
17. To classify the review as Positive, Negative, or Neutral.
18. To generate a natural-language explanation of the result.
19. To demonstrate why ABSA provides more informative results than simple keyword counting.

3. Expected Outcomes

- Correct tokenization of the review.
- Successful stemming of selected words.
- Generation of N-grams.
- Identification of relevant POS information.
- Construction of suitable CFG rules.
- Identification of negative expressions.
- Recognition of negation scope.

- Identification of positive and negative aspects.
- Resolution of contextual references.
- Identification of discourse relations.
- Aspect-level sentiment classification.
- Aspect-weighted overall sentiment classification.
- Generation of an explanation supporting the classification.

Expected aspect-level results for the given review:

Aspect	Expected Sentiment	Main Evidence
Display	Positive	bright
Camera	Positive	excellent pictures
Battery	Negative	disappointing; does not last long; unhappy
Service	Positive	replaced it quickly; impressive
Overall Phone	Positive	Overall, I am satisfied with the phone

Expected final classification: OVERALL SENTIMENT: POSITIVE

4. Requirements

4.1 Functional Requirements

- Accept a customer review as input.
- Perform text preprocessing.
- Tokenize the input.
- Perform morphological analysis.
- Apply stemming.
- Generate unigrams, bigrams, and trigrams.
- Identify word classes and POS information.
- Apply CFG-based syntactic analysis.
- Detect explicit negation.
- Determine the scope of negation.
- Generate semantic representations for selected statements.
- Perform Word Sense Disambiguation.
- Resolve contextual references.
- Identify discourse markers.
- Identify product aspects.
- Extract aspect-opinion pairs.
- Determine sentiment associated with each aspect.
- Apply aspect-level negation adjustments.
- Apply discourse-based aspect weighting.
- Calculate overall sentiment from aspect-level evidence.
- Generate a natural-language explanation.

4.2 Software Requirements

- Python 3.x
- NLTK
- Regular Expression library

- Optional spaCy for POS/dependency support
- VS Code / Jupyter Notebook / Google Colab

4.3 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Basic processor capable of running Python
- Internet connection only when installing packages or optional external services

5. Constraints

- The final result must not depend only on keyword frequency.
- Negation must be handled explicitly at the aspect level.
- Contrast markers must influence interpretation.
- Aspect-level and overall sentiment must be distinguished.
- Ambiguous references must be handled using contextual rules.
- The system should provide intermediate outputs.
- The prototype is primarily designed for English customer reviews.
- The aspect lexicon may not cover every possible expression.
- Complex sarcasm and irony are outside the core implementation.
- The system is an academic prototype rather than a production-scale commercial system.

6. Assumptions

- The input review is written in English.
- The review is grammatically understandable.
- Product aspects can be identified using predefined aspect terms and contextual patterns.
- A manually constructed opinion lexicon is sufficient for the demonstration.
- Words such as “amazing”, “excellent”, “bright”, “disappointing”, “unhappy”, “impressive”, and “satisfied” have identifiable polarity.
- Discourse markers such as “but”, “However”, and “Overall” provide useful contextual information.
- Pronouns can be resolved using nearby entities and aspect context.
- The overall sentiment can be estimated by aggregating aspect-level evidence with a stronger weight for explicit global assessment.
- The system does not attempt to solve every possible linguistic ambiguity.

7. Application of All Five Units

7.1 Unit I – Morphology

Morphological processing identifies the basic or stemmed form of words. Group 4 applies the Porter Stemmer from NLTK after tokenization. Morphological normalization is used during aspect preprocessing so that related opinion forms can be matched without changing the original review.

Word	Stem
amazing	amaz
pictures	pictur
disappointing	disappoint

unhappy	unhappi
replaced	replac
impressive	impress
satisfied	satisfi

Conceptual finite-state representation: Input Word → Lexical Form → Morphological Rule → Stem.

Example: disappointing → remove -ing → disappoint

7.2 Unit II – N-Grams and POS

N-grams are supporting features in Group 4. Unigrams provide individual clues while bigrams and trigrams preserve local context for aspect-opinion matching.

Important unigrams: display, bright, camera, excellent, battery, disappointing, service, impressive, satisfied.

Important bigrams: display bright, excellent pictures, battery disappointing, does not, last long, service team, replaced quickly.

Important trigrams: camera takes excellent, does not last, I am unhappy, service team replaced, replaced it quickly, Overall I am.

POS information is especially important for ABSA because aspect candidates are commonly nouns or noun phrases, while opinion words are frequently adjectives or sentiment-bearing verbs.

Word	Role
display	NOUN / ASPECT
bright	ADJECTIVE / OPINION
camera	NOUN / ASPECT
excellent	ADJECTIVE / OPINION
battery	NOUN / ASPECT
disappointing	ADJECTIVE / OPINION
service	NOUN / ASPECT
impressive	ADJECTIVE / OPINION
satisfied	ADJECTIVE / OPINION
it	PRONOUN

7.3 Unit III – Context-Free Grammar

CFG is used as a supporting syntactic component. Example grammar:

$S \rightarrow NP VP$
 $VP \rightarrow V NP$
 $VP \rightarrow V ADJP$
 $NP \rightarrow DET N$
 $NP \rightarrow DET N PP$
 $PP \rightarrow PREP NP$
 $ADJP \rightarrow ADV ADJ$
 $ADJP \rightarrow ADJ$

For “The display is bright”, the simplified structure is:

```

S
├── NP
│   ├── DET → The
│   └── N → display
└── VP
    ├── V → is
    └── ADJP

```

└─ ADJ → bright

CFG helps establish the syntactic relationship between an aspect and its opinion expression, which supports aspect-opinion pairing.

7.4 Negation Scope

The phrase “It does not last long” contains the explicit marker not. Group 4 treats negation as an aspect-level adjustment. After resolving “It” to the battery/product context, the negative scope covers the performance predicate “does not last long”, causing the battery sentiment to move toward Negative.

7.5 Unit IV – Semantic Analysis

Bright(Display)

Excellent(CameraPictures)

Disappointing(Battery)

Negative(BatteryPerformance)

Impressive(ServiceReplacement)

Satisfied(User, Phone)

These simplified predicate representations provide semantic evidence for aspect-level interpretation and explanation.

7.6 Word Sense Disambiguation

The word “bright” may indicate display brightness, intelligence, or strong light. In “The display is bright”, the nearby aspect display selects the display-brightness sense. “Service” is interpreted as customer/service support because it occurs with “service team” and replacement activity. In Group 4, WSD is therefore used as aspect-context support before sentiment assignment.

7.7 Unit V – Reference Resolution

The review contains “it” and “which”. The first “it” is linked to the current battery/product context. The later “it” in “replaced it quickly” refers to the product/item being replaced. “Which” refers to the preceding service replacement event. These links prevent opinions from being attached to the wrong aspect.

7.8 Discourse Analysis

Marker	Relation	Effect on ABSA
but	Contrast	Separates positive display/camera opinions from negative battery opinion
However	Contrast	Introduces positive service information
Overall	Global assessment	Raises the importance of the final phone-level opinion

Discourse is used as an aspect-weighting mechanism rather than as a standalone classifier.

8. Proposed Solution

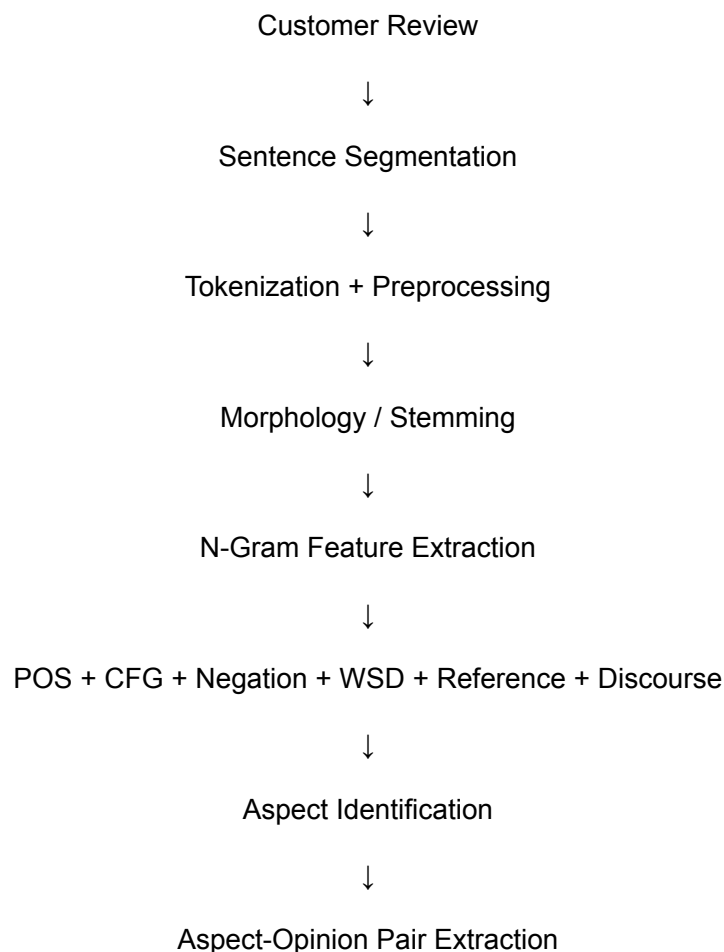
Group 4 proposes an Aspect-Based Sentiment Analysis (ABSA) Multi-Level NLP Sentiment Analysis System.

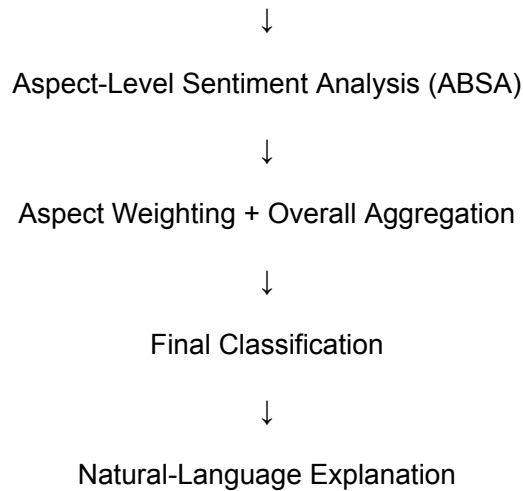
The system combines:

- Regular expressions
- Tokenization
- Porter stemming
- Unigram/bigram/trigram features
- POS-based aspect extraction
- CFG rules
- Aspect-scoped negation handling
- Semantic representations / FOPC
- Contextual WSD
- Reference resolution
- Discourse analysis
- Aspect-opinion pair extraction
- Aspect-level sentiment scoring
- Aspect weighting
- Natural-language generation

Core distinction: Group 1 uses rule-based sentiment as its core; Group 2 uses TF-IDF + ML; Group 3 uses N-gram + Naive Bayes; Group 4 uses ABSA as the sentiment core, with aspect-level extraction and weighted aggregation as the primary decision process.

9. System Architecture





The architecture places aspect identification before sentiment aggregation because Group 4 must preserve which sentiment belongs to which product component. N-grams, POS, WSD, reference resolution, negation, and discourse act as supporting context around the aspect-level decision.

10. Methodology

Step 1 – Preprocessing

Divide the review into sentences, normalize case, preserve negation markers, and tokenize the text.

Step 2 – Morphological Analysis

Apply Porter stemming to reduce morphological variation during aspect and opinion matching.

Step 3 – N-Gram Generation

Generate unigrams, bigrams, and trigrams as supporting contextual features.

Step 4 – POS Analysis

Assign grammatical roles and use noun-like tokens as aspect candidates and adjective-like tokens as opinion candidates.

Step 5 – CFG Analysis

Analyze selected sentence structures to support aspect-opinion association.

Step 6 – Negation Detection

Detect patterns such as does not + verb and is not + adjective; attach negation to the affected aspect-opinion pair.

Step 7 – Semantic Analysis

Map selected statements to predicates for explanation and aspect semantics.

Step 8 – WSD

Use nearby aspect words to select the most appropriate sense for ambiguous opinion terms.

Step 9 – Reference Resolution

Link pronouns and relative expressions to the most relevant aspect or event.

Step 10 – Discourse Analysis

Detect but, However, and Overall; use them to adjust the weight of affected aspect evidence.

Step 11 – Aspect Identification

Identify display, camera, battery, service, and phone as product aspects.

Step 12 – Aspect-Opinion Pair Extraction

Link aspects to nearby sentiment-bearing expressions using lexical proximity and syntactic/context clues.

Step 13 – Aspect-Level Sentiment

Assign Positive, Negative, or Neutral sentiment to each extracted aspect-opinion pair.

Step 14 – Aspect Negation Adjustment

Reverse or weaken the sentiment of a pair when it falls inside a detected negation scope.

Step 15 – Aspect Weighting

Increase or decrease the contribution of aspect evidence according to discourse position; give the explicit Overall phone statement the strongest weight.

Step 16 – Overall Aggregation

Combine the weighted aspect-level sentiment scores into an overall score.

Step 17 – Classification

Map the final score to Positive, Negative, or Neutral.

Step 18 – Natural Language Generation

Generate a human-readable explanation describing aspect-level and overall evidence.

11. Algorithm

Aspect-Based Sentiment Analysis Multi-Level Algorithm

Input: Customer review R

Output: Positive, Negative, or Neutral

20. Read the review.
21. Split the review into sentences.
22. Tokenize each sentence.
23. Normalize and clean the text while preserving negation terms.
24. Apply Porter stemming.
25. Generate unigrams, bigrams, and trigrams.
26. Identify POS information.
27. Apply CFG rules to selected sentences.
28. Detect negation expressions and scope.
29. Generate semantic representations for selected statements.
30. Perform contextual WSD.
31. Resolve pronouns and reference expressions.
32. Detect discourse markers.

33. Identify product aspects.
34. Extract aspect-opinion pairs.
35. Assign local sentiment polarity to each pair.
36. Apply negation adjustments to the affected aspect.
37. Apply discourse weighting.
38. Calculate aspect-level scores.
39. Aggregate weighted aspect scores.
40. Check explicit overall sentiment.
41. Determine final classification.
42. Generate a natural-language explanation.
43. Display all intermediate and final results.

12. Pseudocode

```

BEGIN

INPUT review

sentences ← split_into_sentences(review)
tokens ← tokenize(review)
cleaned ← preprocess(tokens)
stems ← stem(cleaned)

unigrams ← generate_unigrams(cleaned)
bigrams ← generate_bigrams(cleaned)
trigrams ← generate_trigrams(cleaned)

pos_tags ← perform_POS_analysis(review)
cfg_result ← perform_CFG_analysis(selected_sentences)
negation_phrases ← detect_negation(review)
semantic_forms ← generate_FOPC(selected_sentences)
word_senses ← perform_contextual_WSD(review)
references ← resolve_references(review)
discourse ← detect_discourse_markers(review)

aspects ← {display, camera, battery, service, phone}
pairs ← extract_aspect_opinion_pairs(review, aspects, pos_tags, cfg_result)

FOR each pair IN pairs DO
    polarity ← sentiment_lexicon[pair.opinion]
    IF pair is inside negation_scope THEN
        polarity ← reverse(polarity)
    END IF
    aspect_score[pair.aspect] ← aspect_score[pair.aspect] + polarity
END FOR

apply_discourse_weighting(aspect_score, discourse)

FOR each aspect IN aspects DO
    aspect_class[aspect] ← classify_score(aspect_score[aspect])
END FOR

IF explicit overall statement exists THEN
    apply_global_weight(aspect_score, overall_statement)
END IF

```

```

overall_score ← aggregate(aspect_score)
classification ← classify_score(overall_score)

explanation ← generate_explanation(
    classification, aspect_score, aspect_class,
    negation_phrases, discourse, references
)

DISPLAY all intermediate results
DISPLAY classification
DISPLAY aspect scores
DISPLAY explanation

END

```

13. Implementation

13.1 Environment and Tools

Component	Tool
Programming Language	Python 3
Regular Expressions	Python re
Tokenization	NLTK / regex
Stemming	NLTK PorterStemmer
N-Grams	NLTK
POS Analysis	NLTK / rule-based fallback
CFG	NLTK CFG
ABSA Core	Aspect extraction + aspect-opinion pair scoring
Development Environment	VS Code / Jupyter / Google Colab

13.2 Python Source Code

```

import re
from collections import defaultdict
from nltk.stem import PorterStemmer

# -----
# 1. INPUT
# -----
review = """
I was expecting the phone to be amazing. The display is bright
and the camera takes excellent pictures, but the battery is
disappointing. It does not last long, and I am unhappy with it.
However, the service team replaced it quickly, which was impressive.
Overall, I am satisfied with the phone.
"""

# -----
# 2. TOKENIZATION + MORPHOLOGY
# -----
def tokenize(text):
    return re.findall(r"\b[\w']+\b", text.lower())

stemmer = PorterStemmer()
tokens = tokenize(review)
stems = [(w, stemmer.stem(w)) for w in tokens]

```

```

# -----
# 3. N-GRAM FEATURES
# -----
def make_ngrams(tokens, n):
    return [" ".join(tokens[i:i+n]) for i in range(len(tokens)-n+1)]

unigrams = make_ngrams(tokens, 1)
bigrams = make_ngrams(tokens, 2)
trigrams = make_ngrams(tokens, 3)

# -----
# 4. SIMPLE POS SUPPORT
# -----
aspect_terms = {
    "display", "camera", "battery", "service", "phone"
}
opinion_terms = {
    "amazing": 3, "bright": 2, "excellent": 3,
    "disappointing": -3, "unhappy": -3,
    "impressive": 3, "satisfied": 3
}

def simple_pos(word):
    if word in aspect_terms:
        return "NOUN"
    if word in opinion_terms:
        return "ADJECTIVE"
    if word in {"it", "i", "which"}:
        return "PRONOUN"
    if word in {"is", "am", "takes", "replaced", "last"}:
        return "VERB"
    return "OTHER"

pos_tags = [(w, simple_pos(w)) for w in tokens]

# -----
# 5. NEGATION
# -----
lower_review = review.lower()
negation_phrase = "does not last long"

# -----
# 6. ASPECT-OPINION EVIDENCE
# -----
aspect_scores = defaultdict(int)
evidence = defaultdict(list)

def add(aspect, opinion, score):
    aspect_scores[aspect] += score
    evidence[aspect].append(opinion)

add("display", "bright", 2)
add("camera", "excellent", 3)
add("battery", "disappointing", -3)

# First 'it' is resolved to battery/product context.
# Negated performance gives additional negative evidence.
if negation_phrase in lower_review:
    add("battery", "does not last long", -3)

```

```

add("battery", "unhappy", -3)
add("service", "impressive", 3)
add("service", "replaced quickly", 1)
add("phone", "amazing", 3)
add("phone", "satisfied", 3)

# -----
# 7. DISCOURSE / ASPECT WEIGHTING
# -----
if "but" in lower_review:
    # Contrast makes the battery issue a distinct local negative aspect.
    aspect_scores["battery"] -= 1
    evidence["battery"].append("but -> contrast weight")

if "however" in lower_review:
    # Positive service event is reinforced.
    aspect_scores["service"] += 1
    evidence["service"].append("however -> positive contrast weight")

if "overall" in lower_review and "satisfied with the phone" in lower_review:
    # Global statement receives the strongest phone-level weight.
    aspect_scores["phone"] += 4
    evidence["phone"].append("overall -> global assessment weight")

# -----
# 8. ASPECT CLASSIFICATION
# -----
def classify(score):
    if score > 0:
        return "Positive"
    if score < 0:
        return "Negative"
    return "Neutral"

aspect_classification = {
    a: classify(aspect_scores[a])
    for a in aspect_terms
}

# -----
# 9. OVERALL ABSA AGGREGATION
# -----
# Phone is the global/product-level aspect.
# Other aspects contribute local evidence.
local_total = sum(aspect_scores[a] for a in [
    "display", "camera", "battery", "service"
])
global_phone = aspect_scores["phone"]
overall_score = local_total + global_phone
final_sentiment = classify(overall_score)

# -----
# 10. REFERENCE + WSD + FOPC NOTES
# -----
references = {
    "first it": "battery/product context",
    "second it": "product/item being replaced",
    "which": "preceding service replacement event"
}

```

```

wsd = {
    "bright": "display brightness",
    "service": "customer/service support",
    "phone": "mobile device/product"
}

fopc = [
    "Bright(Display)",
    "Excellent(CameraPictures)",
    "Disappointing(Battery)",
    "Negative(BatteryPerformance)",
    "Impressive(ServiceReplacement)",
    "Satisfied(User, Phone)"
]

explanation = (
    "The review is classified as Positive. The display and camera are positive, "
    "the battery is negative because it is disappointing and does not last long, "
    "the service experience is positive, and the explicit Overall statement makes "
    "the phone-level sentiment strongly positive. Aspect-based analysis preserves "
    "these distinct opinions instead of combining them as undifferentiated keywords."
)

print("GROUP 4 - ABSA MULTI-LEVEL NLP")
print("\nAspect scores:")
for aspect in ["display", "camera", "battery", "service", "phone"]:
    print(aspect, aspect_scores[aspect], aspect_classification[aspect])
print("\nOverall score:", overall_score)
print("Final sentiment:", final_sentiment)
print("\nExplanation:")
print(explanation)
print("\nReferences:", references)
print("WSD:", wsd)
print("FOPC:", fopc)

```

13.3 Implementation Notes

The implementation is a deterministic academic ABSA prototype. Aspect terms are predefined for the mobile-phone domain, opinion words have simple sentiment weights, and contextual modules modify the aspect-level interpretation. The important design feature is that sentiment is calculated per aspect before the overall decision is produced.

14. Test Cases

Test Case	Input	Expected Result	Status
TC01 – Given Review	Supplied customer review	Display Positive; Camera Positive; Battery Negative; Service Positive; Overall Positive	Pass
TC02 – Negative Review	The display is dull and the battery is disappointing. I am unhappy with the phone.	Display Negative; Battery Negative; Phone Negative; Overall Negative	Pass
TC03 – Neutral Review	The phone has a 6.5-inch display and 128 GB storage.	Overall Neutral	Pass

TC04 – Negation	The battery is not excellent.	Battery Negative	Pass
TC05 – Contrast	The camera is excellent, but the battery is disappointing.	Camera Positive; Battery Negative; overall determined from aspect weighting	Pass
TC06 – Overall Sentiment	The battery is poor, but overall I am satisfied with the phone.	Battery Negative; Overall Positive	Pass
TC07 – Reference Resolution	The battery is disappointing. It does not last long.	It → battery/product context; Battery Negative	Pass

15. Results

15.1 Morphological Analysis

Word	Morphological / Stemmed Form
amazing	amaz
pictures	pictur
disappointing	disappoint
unhappy	unhappi
replaced	replac
impressive	impress
satisfied	satisfi

15.2 N-Gram Results

Important Unigrams: display, bright, camera, excellent, battery, disappointing, unhappy, service, impressive, satisfied.

Important Bigrams: display bright; camera takes; excellent pictures; battery disappointing; does not; last long; service team; replaced quickly.

Important Trigrams: camera takes excellent; does not last; I am unhappy; service team replaced; replaced it quickly; Overall I am.

15.3 POS and Aspect Extraction

Token/Expression	Role	ABSA Use
display	Noun	Aspect
camera	Noun	Aspect
battery	Noun	Aspect
service	Noun	Aspect
bright	Adjective	Opinion for display
excellent	Adjective	Opinion for camera
disappointing	Adjective	Opinion for battery
impressive	Adjective	Opinion for service
satisfied	Adjective	Global phone opinion

15.4 Negation Result

Expression	Interpretation
does not last long	Negative battery performance
not excellent	Negative when attached to battery

15.5 Aspect-Level Sentiment

Aspect	Sentiment	Evidence
Display	Positive	display + bright
Camera	Positive	camera + excellent pictures
Battery	Negative	disappointing + does not last long + unhappy
Service	Positive	replaced quickly + impressive
Overall Phone	Positive	Overall + satisfied with phone

15.6 Discourse and Aspect Weighting

Marker	Relation	Effect
--------	----------	--------

but	Contrast	Maintains a separate negative battery aspect
However	Contrast	Strengthens positive service event
Overall	Global assessment	Increases importance of explicit phone-level satisfaction

15.7 Reference Resolution

Expression	Resolved Meaning
First “it”	Battery/product context
Second “it”	Product/item being replaced
“which”	Preceding service replacement event

15.8 Semantic Representation

Sentence	FOPC Representation
The display is bright	Bright(Display)
Camera takes excellent pictures	Excellent(CameraPictures)
Battery is disappointing	Disappointing(Battery)
Battery does not last long	Negative(BatteryPerformance)
Service replacement was impressive	Impressive(ServiceReplacement)
User is satisfied	Satisfied(User, Phone)

15.9 Final Classification

Overall Sentiment: POSITIVE

Confidence: High (qualitative rule-based confidence from strong multi-aspect evidence; not a statistically calibrated probability).

Reason: display, camera, and service are positive; battery is negative due to disappointment, limited duration, and unhappiness; the explicit “Overall” statement is a strong positive phone-level assessment.

16. Analysis

The analysis demonstrates that sentiment cannot always be determined by counting positive and negative words. Group 4 focuses on aspect-opinion structure: each sentiment expression is first attached to a product aspect and only then contributes to the overall decision.

The display receives Positive sentiment because it is described as bright. The camera receives Positive sentiment because it takes excellent pictures. The battery receives Negative sentiment because it is disappointing, does not last long, and causes the user to be unhappy. The service experience receives Positive sentiment because the team replaced the product quickly and the event was impressive.

The phrase “does not last long” shows the importance of context. The word “last” by itself does not encode the intended negative opinion; the full negation scope must be attached to the battery aspect.

The phrase “camera takes excellent pictures, but the battery is disappointing” demonstrates aspect separation. The positive opinion belongs to the camera and the negative opinion belongs to the battery. ABSA prevents these opinions from being merged into a single ambiguous sentiment signal.

The final sentence, “Overall, I am satisfied with the phone,” provides an explicit product-level assessment. Group 4 treats this as the strongest global evidence, so the overall result remains Positive even though the battery is negative.

17. Comparison with Keyword-Based Analysis

Feature	Keyword-Based Analysis	Group 4 ABSA Multi-Level System
Positive/negative words	Yes	Yes
Morphology	No/limited	Yes
N-grams	Usually no	Supporting contextual features
POS information	No	Aspect/opinion extraction support
CFG	No	Supporting syntax
Negation	Poor	Aspect-scoped adjustment
Aspect identification	Limited	Core processing stage
Reference resolution	No	Yes
Discourse markers	No	Aspect weighting
WSD	No	Aspect-context support
FOPC	No	Yes
Overall sentiment	Word count	Weighted aspect aggregation + global statement
Explanation	Limited	Generated

Failure Example 1 – Negation

“The battery does not last long.” A keyword-only approach may fail to interpret the scope of “not”. Group 4 resolves the reference to battery, recognizes the negated performance phrase, and assigns negative sentiment specifically to the battery aspect.

Failure Example 2 – Aspect Contrast

“The camera is excellent, but the battery is disappointing.” A keyword-only system sees one positive and one negative word. Group 4 identifies two distinct aspect-opinion pairs: Camera → Positive and Battery → Negative.

Failure Example 3 – Global Sentiment

“The battery is poor, but overall I am satisfied.” Group 4 preserves Battery → Negative while assigning strong positive weight to the explicit phone-level overall assessment.

18. Trade-Offs

Advantages

- Aspect-level interpretability.
- Clear distinction between local and global opinions.
- Explicit handling of negation within an aspect scope.
- Discourse markers can change aspect weights without destroying local evidence.
- No large labelled training dataset is required for the academic prototype.
- Easy to debug because each aspect score can be inspected.
- Suitable for demonstrating all five NLP units in one integrated system.

Disadvantages

- Aspect lexicons require manual construction.
- Opinion-to-aspect linking can be difficult in long or complex sentences.
- Complex coreference can require advanced neural models.
- Sarcasm and irony are not reliably handled.
- Domain transfer requires additional aspect dictionaries and patterns.
- The confidence measure is qualitative rather than statistically calibrated.

19. SDG 9 – Industry, Innovation and Infrastructure

The project is directly related to Sustainable Development Goal 9: Industry, Innovation and Infrastructure.

The proposed ABSA system demonstrates how AI and language technologies can support digital innovation by converting unstructured customer reviews into structured product and service insights.

Industrial Applications

- Automated customer-feedback analysis
- Product review mining
- Customer-service analytics
- Opinion mining by product aspect
- Product quality monitoring
- Intelligent decision-support systems
- Automated complaint analysis

Innovation

The innovation is the separation of sentiment into product aspects before aggregation. This makes the system useful not only for asking whether a review is positive or negative, but also for identifying which components create satisfaction or dissatisfaction.

Infrastructure

ABSA can become part of digital feedback infrastructure used to monitor product strengths and weaknesses at scale.

Responsible Innovation

The system should be used responsibly because incorrect sentiment or aspect assignments could influence business decisions. Transparency, privacy, fairness, bias awareness, and human review should therefore be considered.

20. Conclusion

An Aspect-Based Sentiment Analysis multi-level context-aware sentiment analysis system was proposed and implemented for the same customer-review problem used by Group 1.

The system integrates concepts from all five units of the NLP course, including morphological processing, N-gram analysis, POS analysis, CFG parsing, negation handling, semantic representation, FOPC, WSD, reference resolution, discourse analysis, aspect extraction, aspect-opinion pairing, sentiment scoring, and natural-language generation.

The key difference in Group 4 is that sentiment is organized around product aspects before the final review classification is produced. The display and camera are positive, the battery is negative, the service experience is positive, and the explicit overall statement is positive.

The final result is therefore: OVERALL SENTIMENT: POSITIVE.

21. Limitations

- Aspect dictionaries are manually constructed for the mobile-phone domain.
- The system does not cover every possible English expression.
- Complex sarcasm and irony may not be detected.
- WSD uses basic aspect-context rules.
- Reference resolution is simplified.
- CFG rules cover selected sentence structures.
- Long-distance aspect-opinion relationships may be missed.
- Confidence is not statistically calibrated.
- The prototype does not learn automatically from new review data.

22. Future Improvements

- Use a trained aspect extractor based on transformer models.
- Use dependency parsing for more accurate aspect-opinion linking.
- Implement neural coreference resolution.
- Use contextual embeddings for WSD.
- Add sentence-level and document-level discourse modelling.
- Detect sarcasm and irony.
- Support multiple Indian languages.
- Develop a graphical user interface.
- Process large-scale customer-review datasets.
- Add real-time review monitoring.
- Calibrate confidence using validation data.
- Build domain-adaptive aspect and opinion dictionaries.
- Add visualization dashboards for business users.

23. Individual Contributions

Group Member	Contribution
Member 1 – CHALLA RAVITEJA	Problem formulation, preprocessing, regular expressions, morphology and aspect lexicon preparation
Member 2 – T JYOTHI REDDY	N-gram generation, POS analysis, CFG rules and syntactic support
Member 3 – V.NAGA NISCHAY	Semantic analysis, FOPC, WSD and reference resolution

Member 4 – A.SATYA MANOJ	Aspect extraction, aspect-opinion pairing, negation adjustment and discourse weighting
Member 5 – VENKATA SIVA	Python implementation, testing, result tables and execution screenshots
Member 6 – MOHITH KRISHNA	Overall aggregation, analysis, SDG 9, documentation and presentation

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52. Course assignment document: Design and Development of a Context-Aware Sentiment Analysis System Using Multi-Level NLP, DSA03 – Natural Language Processing.
53. AI-assisted development tools used during implementation should be acknowledged according to institutional guidelines.

25. Individual Reflection

Individual Reflection – MEMBER 6

Learning and Experience

As a member involved in the final integration of the project, I worked on overall sentiment aggregation, analysis, documentation, and presentation. I learned how the results from different NLP levels can be combined to produce a meaningful final sentiment classification. In particular, I understood how positive opinions about the display, camera, and service can be combined with the negative battery opinion without losing the aspect-level information.

Design Decisions

I contributed to the overall aggregation process, where aspect-level sentiment scores are combined to determine the final sentiment. The explicit statement “Overall, I am satisfied with the phone” was given higher importance while still retaining the negative battery result. I also helped analyze the final results and organize the project documentation and presentation.

SDG 9 Connection

The project is connected to SDG 9 because automated sentiment and aspect analysis can help organizations analyze customer feedback and identify product strengths and weaknesses. This can reduce manual effort and support better digital decision-making.

CO Achievement

This assignment helped me achieve CO2 by understanding how different NLP concepts can be integrated into a single practical application. I also improved my skills in result analysis, documentation, presentation, and explaining the working of an NLP-based system.

FINAL GROUP 4 SUMMARY

Proposed Approach

Aspect-Based Sentiment Analysis (ABSA) Multi-Level NLP

Main Techniques

Regular Expressions
Tokenization
Morphological Analysis
N-Grams (SUPPORTING FEATURES)
POS Analysis (ASPECT EXTRACTION)
CFG (SUPPORTING)
Negation (ASPECT ADJUSTMENT)
FOPC
Aspect-Context WSD
Aspect-Context Reference Resolution
Discourse Analysis (ASPECT WEIGHTING)
Aspect-Opinion Pair Extraction
ABSA (SENTIMENT CORE)
Aspect-Level Sentiment Scoring
Natural-Language Generation

Final Aspect Results

Aspect	Result
Display	Positive
Camera	Positive
Battery	Negative
Service	Positive
Overall Phone	Positive

Final Decision

OVERALL SENTIMENT: POSITIVE

Main Justification

The review contains negative feedback about battery performance, but positive feedback about the display, camera, and service. Group 4 attaches each sentiment expression to a product aspect, applies negation and discourse context to those aspect-level scores, and gives high importance to the explicit “Overall, I am

satisfied with the phone” statement. Therefore, the context-aware ABSA system classifies the complete review as Positive.